**EX: 8 Diabetes Prediction using Machine Learning**

**Aim**

To build a machine learning model to predict whether a patient has diabetes using the **Pima Indians Diabetes Dataset** based on medical attributes like glucose level, BMI, age, etc.

**Algorithm**

1. Import required libraries (Pandas, NumPy, Sklearn, Matplotlib).
2. Load the dataset.
3. Perform Exploratory Data Analysis (EDA):
    - View dataset structure
    - Check for missing values
4. Preprocess the data:
    - Replace invalid zero values (Glucose, BMI, etc.)
    - Normalize/scale features if needed
5. Split the dataset into training and testing sets.
6. Train a classification model (e.g., Logistic Regression).
7. Make predictions on test data.
8. Evaluate the model using:
    - Accuracy
    - Confusion Matrix
    - Classification Report
9. Interpret the results.

**Code**

```
# 1. Import Libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# 2. Load Dataset
df = pd.read_csv("diabetes.csv")

# 3. EDA
print(df.head())
print(df.info())
print(df.describe())

# 4. Handling Missing/Invalid Values
cols = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
df[cols] = df[cols].replace(0, np.nan)
df.fillna(df.mean(), inplace=True)

# 5. Splitting Data
X = df.drop('Outcome', axis=1)
y = df['Outcome']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 6. Model Training
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# 7. Prediction
```

```
y_pred = model.predict(X_test)

# 8. Evaluation
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("Accuracy:", accuracy)
print("Confusion Matrix:\n", cm)
print("Classification Report:\n", classification_report(y_test, y_pred))

# 9. Visualization
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

**Result**

- The model successfully classified patients as diabetic or non-diabetic.
- **Accuracy:** ~75% to 80% (may vary slightly based on data split)
- Confusion matrix shows correct and incorrect predictions.
- The model performs reasonably well for medical prediction tasks.